

B.Tech. Degree I Semester Regular Examination November 2023

EE/EC/CS/IT 23-200-0103B INTRODUCTION TO ELECTRONICS DEVICES AND CIRCUITS
(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: To understand the working principle of various semiconductor devices.

CO2: To apply the acquired knowledge to the use of semiconductor devices in various applications.

CO3: To design simple electronic circuits for a given application.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A (Answer ALL questions)

		(5 × 2 = 10)	Marks	BL	CO	PO
I.	(a) Write a short note on doping in semiconductors. What is the need for doping in semiconductors?		2	L1	1	1
	(b) Draw and explain the characteristics of a PN junction diode.		2	L1	1	1
	(c) Explain thermal runaway in transistors.		2	L1	2	1
	(d) A three-stage amplifier has first, second and third stage voltage gains of 10, 15 and 20 respectively. Find the overall voltage gain in decibel.		2	L3	2	2
	(e) What are the benefits of feedback in amplifiers and oscillators?		2	L2	3	1

PART B

(4 × 10 = 40)

II.	Explain the working of a full wave bridge rectifier with suitable waveforms.	10	L1	1	2
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OR

III.	(a) How materials are classified based on their energy band diagram?	6	L2	1	2
	(b) Explain, with suitable waveforms, the working of a capacitor filter.	4	L1	1	1
IV.	(a) Explain the various current components of a BJT. Mention the relation among them.	6	L1	2	1
	(b) Write a short note on how a transistor works as a switch.	4	L1	2	1

OR

V.	Draw and explain the input and output characteristics of BJT in Common Emitter configuration.	10	L1	2	1
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		Marks	BL	CO	PO
VI.	Explain the circuit and working of a Common Emitter RC coupled amplifier and its frequency response behavior.	10	1	2	1
OR					
VII. (a)	Describe the working of a Common Source FET amplifier.	6	L1	2	1
(b)	Explain the four types of feedback amplifiers with block diagram.	4	L2	2	1
VIII.	Explain the circuit and working of an RC phase shift oscillator.	10	L1	3	1
OR					
IX.	How an astable multivibrator works? Explain with a neat circuit diagram.	10	L2	3	1

Bloom's Taxonomy Levels

L1 - 73.3%, L2 - 24.5%, L3 - 2.2%.
